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Oefeningenreeks 2E nr. 4.22

$$\lim_{x\to\infty}\left(\frac{x^2+2x-1}{x^2+x-2}\right)^{x-2}$$
$$=\lim_{x\to\infty}\left(\frac{x^2+2x-x+x-1-1+1}{x^2+x-2}\right)^{x-2}$$
$$=\lim_{x\to\infty}\left(\frac{x^2+x-2+x+1}{x^2+x-2}\right)^{x-2}$$
$$=\lim_{x\to\infty}\left(\left(1+\frac{x+1}{x^2+x-2}\right)^{\frac{x^2+x-2}{x+1}}\right)^{\frac{(x-2)(x+1)}{x^2+x-2}}$$
$$=e^{\lim\limits_{x\to\infty}\frac{(x-2)(x+1)}{x^2+x-2}}$$
$$=e^{\lim\limits_{x\to\infty}\frac{x^2}{x^2}}$$
$$=e$$

References